

Peer Feedback Log — KijaniKiosk Capstone

1. Kubernetes Connectivity and Context Misconfiguration (High Severity)

Issue:

Initial deployments failed because `kubectl` inside Jenkins was pointing to an incorrect or inaccessible Kubernetes API server (`127.0.0.1`), leading to authentication and connection errors.

Impact:

Pipeline could not complete the staging deployment stage, causing repeated CI/CD failures.

Root Cause:

Mismatch between host kubeconfig and container runtime environment. The kubeconfig context was not compatible with Jenkins running inside a Docker container.

Resolution Attempt:

- Recreated Kind cluster (`kijani-cluster`)
- Exported kubeconfig using `kind export kubeconfig`
- Verified cluster health using `kubectl get nodes`

Status:

Partially resolved — cluster connectivity works on host but still requires careful handling inside Jenkins container.

2. Jenkins Container Tooling Instability (High Severity)

Issue:

Jenkins pipeline failed at different stages due to missing tools inside the container (`docker`, `kubectl`), and inconsistent runtime environment across restarts.

Impact:

Pipeline stages such as build and deploy failed unpredictably depending on container state.

Root Cause:

Jenkins container was not built as a hardened CI image and relied on manual runtime installation of dependencies.

Resolution Attempt:

- Installed Docker CLI inside Jenkins container
- Mounted Docker socket from host (`/var/run/docker.sock`)
- Installed `kubectl` manually

Status:

Improved but not fully stable; environment still depends on manual setup after container recreation.

3. Missing Kubernetes Namespaces (Medium Severity)

Issue:

Deployment stages failed due to missing namespaces (`kijani-staging`, `production`) referenced in the Jenkins pipeline.

Impact:

`kubectl apply` commands failed during staging deployment stage.

Root Cause:

Infrastructure prerequisites (namespaces) were not pre-provisioned in cluster setup.

Resolution Attempt:

- Created namespaces manually using `kubectl create namespace`
- Updated Jenkins pipeline to auto-create namespaces if missing

Status:

Resolved — pipeline now handles missing namespaces more gracefully.

4. Jenkins Pipeline SCM and Branch Mismatch (Medium Severity)

Issue:

Initial pipeline configuration failed to fetch repository due to incorrect branch reference (`master` vs `main`), causing SCM checkout failure.

Impact:

Pipeline failed before execution of any build stage.

Root Cause:

Mismatch between Jenkins SCM configuration and actual Git repository default branch.

Resolution Attempt:

- Verified repository branch structure
- Adjusted Jenkins job configuration to correct branch reference

Status:

Resolved after correcting SCM configuration.

5. Docker Build and Permission Issues in CI Environment (High Severity)

Issue:

Docker build initially failed inside Jenkins due to:

- Missing Docker CLI
- Permission denied when accessing Docker daemon socket

Impact:

Build stage consistently failed even when code and pipeline were correct.

Root Cause:

Jenkins container was not properly configured with Docker socket permissions and CLI tooling.

Resolution Attempt:

- Mounted `/var/run/docker.sock`
- Installed Docker CLI inside Jenkins container
- Adjusted container run permissions (`--user root` during setup phase)

Status:

Partially resolved — works in current session but requires careful setup persistence strategy.

Overall Reflection

The system evolved from:

- No CI/CD pipeline → working Jenkins pipeline structure
- No Kubernetes integration → functioning Kind cluster deployment
- No containerized CI environment → Docker-enabled Jenkins setup

However, the main limitation remains:

The CI environment is still manually assembled rather than fully reproducible as Infrastructure-as-Code.

Suggested Improvements (Future Work)

- Build a custom Jenkins Docker image with preinstalled Docker + `kubectl`
- Fully automate Kind cluster provisioning using Terraform or scripts

- Add health-based smoke tests instead of placeholder curl commands
- Add Prometheus-based monitoring for deployment verification
- Improve kubeconfig handling for containerized CI environments